Software Quality Assurance

Quality concepts · ISO 9000 · SEI CMM · PSP · Six Sigma

Organization of this lecture

1. **Quality concepts**
   What quality means for software; quality factors
2. **Quality systems**
   QC, QA, TQM and how they evolved
3. **ISO 9000**
   Standards, certification and shortcomings
4. **SEI CMM**
   Five maturity levels and key process areas
5. **PSP & Six Sigma**
   Personal process and defect reduction

1

What is software quality?

From 'fitness of purpose' to multiple quality factors

Traditional view: fitness of purpose

✓

The traditional definition

- A quality product does exactly what users want it to do
- For software: it satisfies the requirements in the SRS document
- Works well for products like a car, table fan, food mixer or microwave oven

✗

Why it falls short for software

- Functionally correct software with an almost unusable interface is not a quality product
- Software that does everything users want but has incomprehensible, unmaintainable code is not a quality product
- So software needs a broader, multi-factor definition

Modern view: software quality factors

C

Correctness

SRS requirements implemented correctly; accurate results

R

Reliability

Works consistently without failure over time

E

Efficiency

Makes good use of time and resources (CPU, memory)

P

Portability

Runs easily on other OSs, machines and software

U

Usability

Both novice and expert users can use functions easily

R

Reusability

Modules can be reused to build new products

M

Maintainability

Easy to fix errors, add or modify functions

Remember

Fitness of purpose covers only correctness. The modern view adds the other six.

2

Quality management systems

Structure, activities and the evolution from inspection to TQM

What is a quality system?

The principal methodology an organization uses to make sure its products have the desired quality.

1

It consists of

- Managerial structure
- Individual responsibilities
- Responsibility of the organization as a whole

2

Quality department activities

- Auditing projects
- Developing standards, procedures and guidelines
- Reporting to top management
- Reviewing the quality system itself

3

Success conditions

- Independent quality department
- Top-management support, or staff won't take it seriously
- Well documented, or quality varies from product to product

Evolution of quality systems

1

Product inspection

Check finished products; reject the defective ones

2

Quality control

Reject defects and find their causes; statistical sampling

3

Quality assurance

Good process, followed rigorously → good product

4

TQM

Continuous process improvement through measurement

The paradigm shift: from product assurance to process assurance

QC vs QA vs TQM at a glance

Aspect	Quality control	Quality assurance	TQM
Focus	Product	Process	Continuous process improvement
Approach	Detect defects, find causes	Prevent defects with a defined process	Measure and redesign processes
Nature	Corrective	Preventive	Preventive and improving
Key idea	Statistical sampling instead of testing everything	Good process → good product	Optimise, not only document

3

ISO 9000

International standards for quality systems

ISO 9000: overview and the three standards

- Published by ISO, a consortium of 63 countries, in 1987
- Guidelines for maintaining a quality system, not the product itself
- Covers responsibilities, reporting, procedures, processes and resources
- Serves as a reference for contracts between independent parties
- **Premise: if a proper process is followed, good products follow**

ISO 9001 Design, development, production and servicing. Applies to software.

ISO 9002 Production only, no design (e.g. steel, car manufacturing).

ISO 9003 Installation and testing only.

Why ISO 9000 is hard to apply to software

M

Manufacturing (e.g. car plant)

- Product is tangible: you can see engine and doors being fitted
- Status of the product is known at any time
- Consumes large amounts of raw material (iron ore, coal, limestone)

S

Software development

- Software is intangible, so it is hard to control and track
- The only raw material consumed is data
- ISO's raw-material clauses are not relevant

ISO released ISO 9000 Part-3 in 1991 to interpret the standard for software, but official guidance is still considered inadequate.

Why get ISO 9000 certification?

1

Win contracts

Many international software contracts require ISO 9000 certification

2

Documented process

Requires a well-documented production process; leads to repeatable, higher-quality software

3

Efficiency

Makes development focused, efficient and cost-effective

4

Find weaknesses

Points out the organization's weaknesses and recommends remedial action

5

Foundation for TQM

Sets the basic framework for an optimal process and TQM

How to get ISO 9000 certification

1

Application

Organization applies to an ISO 9000 registrar

2

Pre-assessment

Registrar makes a rough assessment

3

Document review

Registrar reviews process documents and suggests improvements

4

Compliance audit

Registrar checks suggestions were complied with

5

Registration

Certificate awarded after all phases

6

Surveillance

Registrar keeps monitoring periodically

Rule: The certificate may be used in corporate advertising, never in product advertising. ISO certifies the process, not a product; misuse risks withdrawal.

Summary of ISO 9001 requirements

Clause	Requirement	In software terms
4.1	Management responsibility	Effective quality policy; independent quality system, audited periodically
4.2–4.3	Quality system, contract review	Documented system; review contracts before accepting
4.4	Design control	Verify design inputs and outputs; control design changes and coding
4.5	Document control	Approve, issue and change documents via configuration management
4.6–4.7	Purchasing, purchaser-supplied	Check bought-in and client-provided software
4.8–4.9	Product identification, process control	Configuration management; quality plan
4.10–4.11	Inspection and testing	Unit, integration, system testing; keep records; calibrate tools
4.13–4.14	Nonconforming product, corrective action	Keep faulty code out of releases; fix root causes
4.15–4.18	Handling, audits, training	Storage and delivery; quality audits; identify training needs

Shortcomings of ISO 9001 certification

1

No process quality guarantee

Requires a process to be followed but doesn't ensure the process is good, or say how to define one

2

Not fool-proof

No international accreditation agency; norms vary between agencies and registrars

3

Downplays domain expertise

Assumes any engineer is as effective as any other once a process exists

4

Software is creative

Unlike manufacturing, individual skill and experience matter a lot

5

No automatic improvement

Doesn't automatically lead to continuous improvement or TQM

6

Management-focused

Ignores software techniques: configuration management, reviews, release builds

4

SEI Capability Maturity Model

Five levels of software process maturity

SEI CMM: origin and uses

- Developed by the Software Engineering Institute (SEI), Carnegie Mellon University, USA
- Built to help the U.S. Department of Defense choose software contractors
- Rates a contractor's software process maturity into levels
- Predicts the most likely outcome of the organization's next project
- **Levels are ordered: each one is the foundation for the next**

E

Capability evaluation (external)

Assesses an organization's process capability. Helps select a contractor and indicates likely performance.

I

Software process assessment (internal)

An organization assesses its own process and finds ways to improve. For internal use only.

CMM is used in **two primary ways**:

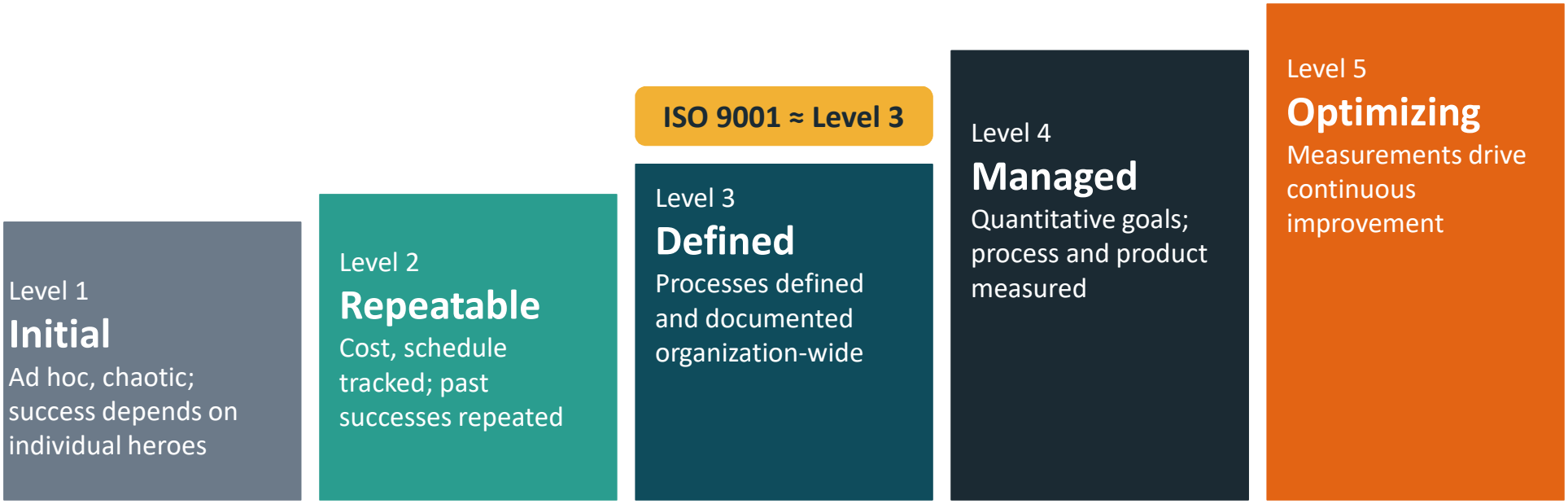
1. Capability Evaluation (Software Capability Evaluation)

- **Purpose:** External assessment to evaluate the process capability of a supplier, vendor, or contractor.
- **Goal:** Predict the likely quality, cost, and schedule performance of a contractor before awarding them a project.
- **Usage:** Conducted externally to help **select the right contractor**.

2. Software Process Assessment

- **Purpose:** Internal review conducted by or for an organization itself.
- **Goal:** Identify the strengths and weaknesses of the organization's current software processes and build a roadmap for growth.
- **Usage:** Purely **for internal process improvement**, helping teams move up the five CMM maturity levels (Initial, Repeatable, Defined, Managed, and Optimizing).

The five maturity levels



Key process areas (KPAs) for each level

A KPA shows where an organization at the previous level must focus to reach the next level.

Level 2: Repeatable	Level 3: Defined	Level 4: Managed	Level 5: Optimizing
• Software project planning (size, cost, schedule) • Project monitoring • Configuration management • Subcontract management	• Process definition and documentation • Reviews • Training program	• Quantitative measurements • Process management	• Defect prevention • Technology change management • Process change management

ISO 9001 vs SEI CMM

Aspect	ISO 9001	SEI CMM
Awarded by	International standards body	Assessment, mainly for internal use
Official use	Can be quoted in official documents	Purely internal use
Scope	Generic: any industry	Specifically for software
Goal	Quality assurance	Goes beyond QA; aims at TQM
Levels	Single certification (≈ CMM Level 3)	Five maturity levels
Improvement path	No guidance on how to improve	KPAs give a step-by-step path

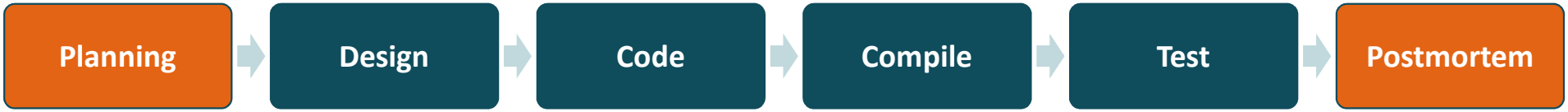
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PSP and Six Sigma

Quality at the individual level and in numbers

Personal Software Process (PSP)

Based on Watts Humphrey's work: a scaled-down industrial process that helps an individual engineer measure and improve how they work.



Log the time spent in every phase · compare planned vs actual in the project plan summary

0

PSP 0

Personal measurement; basic size measures

1

PSP 1

Personal planning; time and schedule

2

PSP 2

Personal quality management; design and code reviews

3

PSP 3

Personal process evolution

Six Sigma

3.4

defects per million opportunities at 6σ

230

defects per million at 5σ

6,210

defects per million at 4σ

1

DMAIC

- Define
- Measure
- Analyze
- Improve
- Control
- Improves an existing process

2

DMADV

- Define
- Measure
- Analyze
- Design
- Verify
- Designs a new process

Summary: key takeaways

1

Quality

Means more than meeting the SRS: seven quality factors

2

Evolution

Inspection → QC → QA → TQM; the focus moved from product to process

3

ISO 9000

Good process → good products; 9001 applies to software

4

SEI CMM

Built for software; five maturity levels with KPAs

5

PSP & Six Sigma

Individual discipline; 3.4 defects per million

6

Current trend

Reduce delivery time and adopt Six Sigma